
SHRINKAGE BASELINES FOR GRPO ON GSM8K: WHEN JAMES–STEIN COLLAPSES, AND WHAT IT TAKES TO MEASURE IT

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Code and logs: https://github.com/hyang150/GRPO_JS_Eva

ABSTRACT

GRPO estimates each prompt’s baseline from the N completions sampled for that prompt alone; with small N this mean is noisy. A given specification proposes replacing it with a James–Stein estimator that shrinks each group mean toward the batch mean, with the shrinkage constant fixed by assuming unit per-sample reward variance. We implement the specification, five alternative baselines behind one registry, and measure them on Qwen2.5-0.5B-Instruct / GSM8K. Three results. (i) Applied to raw 0/1 rewards the specified estimator collapses onto the global batch mean: its assumed variance overstates the true one by at least $4\times$, the collapse is monotone in the number of prompts K , and at $N = 2$ it is an algebraic identity for every $K > 6$. In three-seed training runs it discards the group structure on 100% of steps. (ii) Establishing the specification’s own $\sigma^2 \approx 1$ precondition restores Stein’s behaviour: the shrinkage factor follows the closed form $N\tau^2/(1 + N\tau^2)$ to within 0.02, and the baseline’s mean squared error falls by 16–54% in simulation. (iii) None of this is resolvable as end-to-end accuracy at this scale. The two arms that are provably the same algorithm at $K = 32, N = 2$ finish 0.5, 6.5 and 1.5 points apart across seeds, against a literature effect of 0.65–1.5 points; the per-step gradient noise-to-signal ratio is 58–182 for every baseline alike. A paired gradient-variance measurement that an earlier draft reported as significant is withdrawn after a projection bug and an off-policy sampler were found, and was not repeated within the time budget. We also show that the specification’s optional division by the within-group standard deviation multiplies degenerate-group advantages by 10^4 under any shrinkage baseline.

1 INTRODUCTION

Group Relative Policy Optimization (Shao et al., 2024) removes the critic from PPO (Schulman et al., 2017) by sampling N completions per prompt and using their mean reward as the baseline. With verifiable 0/1 rewards and small N that baseline is a coarse estimate: at $N = 2$ it takes three values, and a group that is entirely right or entirely wrong carries no signal at all. The specification this work was given (GRPO.md, reproduced in Appendix A) proposes a remedy from classical statistics: treat the K group means in a batch as K normal means with a common known variance $V = 1/N$ and shrink them toward the grand mean with the positive-part James–Stein estimator (James & Stein, 1961). Stein’s result then promises a lower total mean squared error, from which the specification infers lower advantage variance, more stable training and support for smaller N .

We set out to test that chain of inference end to end. The contributions are:

- A registry of six baselines (Sec. 4) with a literal transcription of the specification asserted against the implementation to 10^{-12} , an adapter to verl’s advantage-estimator interface, and 162 tests. The specification’s own pseudocode is found to crash on two of the edge cases its prose claims to handle (Appendix B).

- An analysis (Sec. 3) of when the specified estimator degenerates into the global mean: the large- K limit of its shrinkage term, the $N = 2$ identity, and the closed form the shrinkage factor follows once the precondition is honoured.
- Simulation with known ground truth (Sec. 5.1) and three-seed GRPO training at two group shapes with $K \cdot N = 64$ (Sec. 5.3), including a same-algorithm replicate pair that measures the protocol’s noise floor directly.
- A corrections record (Appendix C): an off-policy sampler, a completion-mask defect and a gradient-projection bug, all found in review, all fixed, and the withdrawal of the result they contaminated.

The headline is negative in two senses. The specification as written does not work on 0/1 rewards, and the corrected version, while it does what Stein’s theorem says it should to the baseline, produces no accuracy effect that this scale of experiment can resolve.

2 BACKGROUND

GRPO. For prompt x the policy samples N completions $y_1, \dots, y_N$ and receives rewards r_i . GRPO forms per-sample advantages $A_i = r_i - b$ with b the group mean, optionally divided by the group standard deviation, and applies the PPO clipped surrogate with a KL penalty toward a reference policy. Subtracting a baseline that does not depend on the sample leaves the policy gradient unbiased and lowers its variance; the group mean is a per-prompt Monte Carlo estimate of the value function, which is what makes GRPO work without a critic. It is not independent of r_i — it contains it — but for the plain mean this only scales the gradient by $(1 - 1/N)$; the leave-one-out mean of RLOO (Ahmadian et al., 2024) removes even that.

The specification. With group means $X_k = \frac{1}{N} \sum_i r_i^{(k)}$, grand mean $\bar{X}$, $S = \sum_k (X_k - \bar{X})^2$ and $c = V(K - 3)$,

$$\tilde{\mu}_k = \bar{X} + \text{shrink} \cdot (X_k - \bar{X}), \quad \text{shrink} = \max\left(0, 1 - \frac{c}{S}\right), \quad (1)$$

and advantages are $A_i^{(k)} = r_i^{(k)} - \tilde{\mu}_k$. The specification sets $V = 1/N$, which is correct if the per-sample reward variance σ^2 is 1, and states in its theory section that rewards should first be rescaled to make it so. Its pseudocode takes the raw reward matrix.

3 ANALYSIS

3.1 THE PRECONDITION, AND WHAT K IS

On GSM8K the reward is binary, so $\sigma_k^2 = p_k(1 - p_k) \leq \frac{1}{4}$: at least $4\times$ smaller than assumed, heteroscedastic across prompts, and exactly zero for a group that is entirely right or entirely wrong (25–28% of groups at $N = 8$ and 68–70% at $N = 2$ in our runs). Overstating V overstates c , which drives shrink toward zero.

K is the number of prompts whose rewards are on hand when advantages are formed — the rollout batch, not the backward micro-batch. Stein’s phenomenon needs $K \geq 3$ for a fixed shrinkage centre and $K \geq 4$ when the centre is the estimated grand mean, hence $K - 3$. This work runs $K = 8$ and 32; Zeng et al. (2025) run 64; production runs use hundreds to thousands.

3.2 THE LARGE- K LIMIT

Since $\mathbb{E}[S] \approx (K - 1) \text{Var}(X_k)$ and $c = V(K - 3)$, with τ^2 the variance of the true per-prompt pass rates,

$$\frac{c}{S} \xrightarrow{K \rightarrow \infty} \frac{V_{\text{assumed}}}{\text{Var}(X_k)} = \frac{V_{\text{assumed}}}{\tau^2 + \sigma^2/N}, \quad (2)$$

a constant. K does not make the shrinkage stronger; it makes S less noisy, so the estimator converges to whatever this ratio says.

name	baseline $b[k, i]$	role
vanilla	group mean	GRPO, the incumbent
rloo	leave-one-out group mean	unbiased variance-reduction control
global	batch mean	ablation: what over-shrinkage degenerates to
js_fixed	Eq. (1), $V = 1/N$	the specification as written
js_pooled	Eq. (1), V estimated	the specification on r/σ_{pooled} , scaled back
js_loo	two-level leave-one-out shrinkage	Zeng et al. (2025)

Table 1: Baselines. Without `global` the comparison is uninterpretable: “James–Stein helps” cannot be told apart from “any baseline other than the group mean helps”.

If the precondition held, $V_{\text{assumed}} = 1/N$ is exact and the limit is $1/(1 + N\tau^2) < 1$ for every N and τ^2 : the clamp never fires and

$$\text{shrink} \xrightarrow{K \rightarrow \infty} \frac{N\tau^2}{1 + N\tau^2}, \quad (3)$$

the empirical-Bayes posterior weight — mild shrinkage that weakens as N grows or as prompts spread apart.

If it does not, as for 0/1 rewards, the ratio is at least $(1/N)/(\tau^2 + 1/(4N))$. For the reward distribution used in Sec. 5.1 it is 1.69 at $N = 8$ and 3.57 at $N = 2$. Above 1 the positive-part clamp fires and the estimator *is* the global mean. Larger K makes this certain rather than occasional: the specified method gets worse as the batch holds more prompts, which is the regime GRPO is deployed in.

3.3 THE $N = 2$ IDENTITY

Group means of 0/1 rewards lie in $[0, 1]$, so $S \leq K/4$ for any K . With $c = (K - 3)/N$ the clamp is guaranteed whenever $(K - 3)/N > K/4$, i.e. $K > 3/(1 - N/4)$: $K > 6$ at $N = 2$, $K > 12$ at $N = 3$, and never guaranteed for $N \geq 4$, where it needs S to be small, which Eq. (2) supplies as K grows. At $N = 2$, the regime the specification is motivated by, the specified estimator therefore equals the global mean for every batch of more than six prompts, independently of the rewards. Sec. 5.3 turns this into a measurement.

3.4 WHAT DOES NOT TRANSFER EVEN UNDER THE PRECONDITION

Three things. Bernoulli rewards are heteroscedastic, so a common known V is an approximation established by pooling; the per-group standard deviation cannot be used because it is zero for a third to two thirds of groups. The shrunk baseline depends on r_i itself through $\bar{X}$ and S , so the policy gradient is biased — not by the constant $(1 - 1/N)$ factor of plain GRPO but nonlinearly; only leave-one-out constructions keep it unbiased. And a degenerate group receives a non-zero advantage by design, which pushes all-correct prompts up and all-wrong prompts down in whichever direction the batch happens to lean: noise traded for bias, not signal created.

3.5 THE STANDARD-DEVIATION OPTION

The specification offers “divide by the within-group standard deviation afterwards” as an option. Under plain GRPO that is safe on a degenerate group only because the advantage is exactly zero there, so $0/(0 + 10^{-4}) = 0$. Every shrinkage baseline gives such a group a non-zero advantage — that is the point of the method — so the same expression multiplies it by 10^4 . Measured on GSM8K at $K = 8$, $N = 8$ with 38% degenerate groups: $\text{Var}(A) = 1.07 \times 10^5$ and gradient norm 1.7×10^3 ; leaving zero-spread groups unscaled gives 0.57 and 1.2. All experiments below use no standard-deviation scaling, which is also the choice Dr.GRPO (Liu et al., 2025) argues for on other grounds.

N	degenerate	global	js_fixed	js_pooled	js_loo	js_pooled shrink (Eq. 3)
2	64.5%	-39.6%	-39.6%	-54.4%	-51.3%	0.386 (0.370)
4	36.7%	+17.2%	+17.2%	-40.8%	-40.0%	0.547 (0.541)
8	18.7%	+131.1%	+131.0%	-26.7%	-27.0%	0.705 (0.702)
16	8.9%	+355.5%	+258.8%	-15.8%	-16.0%	0.827 (0.825)

Table 2: MSE of the baseline against the true p_k relative to vanilla, $K = 32$. `rloo` equals vanilla exactly: leave-one-out changes the baseline’s independence, not its point estimate. The last column gives `js_pooled`’s measured shrinkage factor with the prediction of Eq. (3) in parentheses.

K	js_fixed shrink	js_fixed MSE	global MSE	js_pooled MSE
4	0.333	+1.0%	+99.1%	-9.3%
8	0.058	+84.3%	+116.4%	-18.7%
16	0.006	+121.7%	+125.5%	-24.4%
32	0.000	+130.2%	+130.5%	-26.8%
64	0.000	+133.6%	+133.6%	-28.1%

Table 3: Sweeping K at $N = 8$. By $K = 64$ `js_fixed` and `global` agree to every digit. At $K = 4$ the specified method is roughly neutral, so a small- K experiment would show nothing wrong.

4 ESTIMATORS

Table 1 lists the six. `js_pooled` is the specification’s ten steps applied after dividing the rewards by the pooled within-group standard deviation $\sqrt{\text{mean}_k s_k^2}$ and scaling back; the identity is pinned by a test to 3×10^{-16} . We read it as the specification with its own precondition honoured, not as a different method. `js_loo` follows Zeng et al. (2025), Sec. 3.3, eq. 10 and 13–16, with both the group mean and the shrinkage target leave-one-out. The registry mirrors verl’s advantage-estimator interface so any arm can be dropped into that framework through a thin adapter. The training loop fixes everything that could confound a baseline comparison: no KL term ($\beta = 0$, no reference model), no standard-deviation scaling, constant loss normalisation so completion length does not reweight the update.

5 EXPERIMENTS

5.1 SIMULATION WITH KNOWN GROUND TRUTH

Ground truth is unavailable on GSM8K, so the baseline’s error is measured in simulation: $p_k \sim \text{Beta}(1.2, 2.2)$ (mean 0.35, matching the model’s observed per-prompt spread; $\tau^2/\sigma^2 = 0.294$), rewards $\sim \text{Bernoulli}(p_k)$, 2000 trials at $K = 32$ and 3000 at each K of the sweep.

Table 2 gives the N sweep. `js_fixed` is `global` for $N \leq 4$ and within 0.1% of it at $N = 8$: it discards the group structure entirely, and for $N \geq 4$ it is worse than plain GRPO. Estimating V fixes it, the gain growing as N shrinks, which is the regime the method claims; and the measured shrinkage follows Eq. (3) to within 0.02. Table 3 shows the collapse deepening monotonically in K , as Eq. (2) predicts. Figure 1 draws both.

5.2 GRPO ON GSM8K

Qwen2.5-0.5B-Instruct (Qwen Team, 2024), full fine-tune, GSM8K (Cobbe et al., 2021) with a rule-based correctness reward, 512 new tokens, bf16 with gradient checkpointing, learning rate 10^{-6} , PPO clip 0.2, one inner update per rollout. Development on one RTX 5080 (16 GB; 11.0 GiB peak, ~ 25 s per step at $K=8, N=8$), the three-seed sweeps on two RTX 5090s. Every evaluation scores the same 200 test problems greedily in the same order, so checkpoints are compared with McNemar’s test rather than by overlapping binomial intervals.

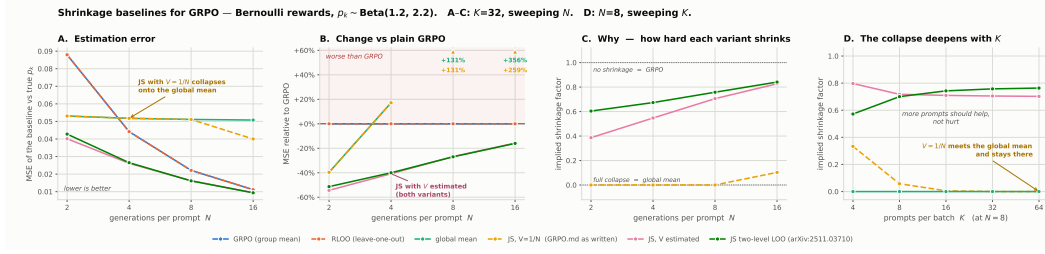


Figure 1: A–C: sweeping N at $K = 32$; D: sweeping K at $N = 8$. `js_fixed` (dashed) lies on global; estimating V recovers the intended behaviour.

$K \times N$	arm	final, per seed	mean $\pm$ sem	vs vanilla	p	shrink	collapsed
8 $\times$ 8	vanilla	44.5 / 42.0 / 47.5	44.7 $\pm$ 1.6			1.000	0
8 $\times$ 8	js_pooled	44.0 / 45.5 / 47.0	45.5 $\pm$ 0.9	+0.8 $\pm$ 1.3	0.60	0.85	0–1
8 $\times$ 8	js_fixed	44.0 / 47.5 / 46.0	45.8 $\pm$ 1.0	+1.2 $\pm$ 2.2	0.65	0.21	38–44
8 $\times$ 8	global	48.5 / 44.0 / 45.0	45.8 $\pm$ 1.4	+1.2 $\pm$ 1.9	0.61	0.000	150
32 $\times$ 2	vanilla	46.0 / 45.5 / 44.5	45.3 $\pm$ 0.4			1.000	0
32 $\times$ 2	js_pooled	41.0 / 46.0 / 47.5	44.8 $\pm$ 2.0	−0.5 $\pm$ 2.4	0.85	0.56	0–1
32 $\times$ 2	js_fixed	43.5 / 42.0 / 41.5	42.3 $\pm$ 0.6	−3.0 $\pm$ 0.3	0.009	0.000	150
32 $\times$ 2	global	44.0 / 48.5 / 43.0	45.2 $\pm$ 1.7	−0.2 $\pm$ 1.6	0.93	0.000	150

Table 4: Final test accuracy (200 problems, greedy) after 150 steps, seeds 0–2, corrected sampler. “vs vanilla” is the within-seed paired difference, p a paired t -test with two degrees of freedom, shrink the run mean of the implied shrinkage factor, “collapsed” the number of steps at exactly 0 (range over seeds). Every run starts from the same checkpoint, so accuracy at step 0 is 45.0% for all 24.

Two engineering points affect the numbers. Gradient checkpointing sets `use_cache=False`, which silently makes generation quadratic in length; toggling it off around the rollout halved the step time. And pure bf16 at 10^{-6} rounds most of the update away — a weight of 0.0145 has a ULP of 1.1×10^{-4} while an AdamW step moves it by $\sim 10^{-6}$; measured, 2.3% of sampled weights change per step. The fp32-master-weight recipe does not fit 16 GB at this batch, so all arms share the bf16 handicap.

5.3 THREE SEEDS, TWO GROUP SHAPES

Four arms at each of two shapes with $K \cdot N = 64$, so every arm sees the same number of completions per step; every arm of a seed walks the same prompt stream. Table 4 gives the result.

At $K = 8$, $N = 8$ nothing is resolved. Every arm is within 1.2 points of `vanilla`, every sem exceeds the difference, every p is about 0.6. The per-seed columns show why: the same arm moves 3–5 points between seeds. Seed 0 alone would have said `global` beats `vanilla` by 4.0 points (McNemar $p = 0.039$); seeds 1 and 2 put it at +2.0 and −2.5.

At $K = 32$, $N = 2$ the one significant row is the noise floor. `js_fixed` at -3.0 ± 0.3 ($p = 0.009$) reads as a finding. But by Sec. 3.3 it is `global` at this shape: its logged shrinkage is 0 on all 150 steps of all three seeds, and its step-1 record agrees with `global`’s to the bit before the two processes diverge at step 2, because sampled generation is not reproducible on GPU. The two arms are therefore three same-seed replicates of one algorithm (Table 5): they end 0.5, 6.5 and 1.5 points apart. Pooling them as the six runs they are gives -1.6 ± 1.0 against `vanilla` and the significance is gone. A t -test with two degrees of freedom on three differences that happen to land within 0.5 of each other produces $p < 0.01$ from nothing.

The effect reported for this setting is 0.65–1.5 points (Zeng et al. (2025), Table 2: JS minus GRPO at $N = 8/4/2$ on this model and task, $K = 64$, 500 steps, five seeds). The noise floor is 2–5 $\times$ the effect; resolving a one-point effect at this spread needs on the order of 20–40 seeds per arm.

seed	global	js_fixed	difference
0	44.0%	43.5%	+0.5
1	48.5%	42.0%	+6.5
2	43.0%	41.5%	+1.5

Table 5: The same algorithm run twice per seed ($K = 32$, $N = 2$). The spread is the run-to-run term of this protocol. An earlier two-run estimate under the 1.1 sampler was 3.0 points.

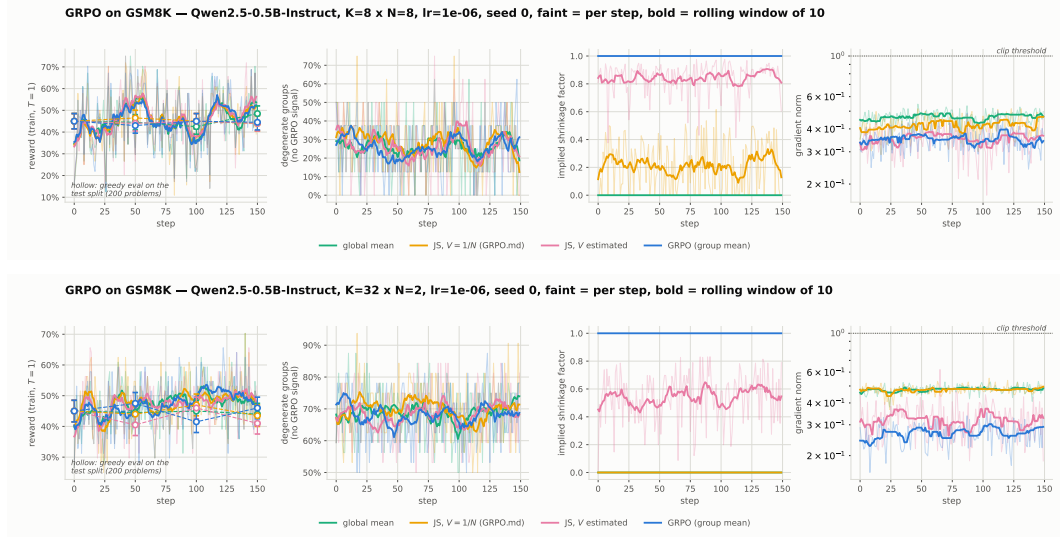


Figure 2: Seed 0 at $K=8$, $N=8$ (top) and $K=32$, $N=2$ (bottom). Reward and degenerate-group panels overlap; the shrinkage panel separates the arms, and at $N = 2$ `js_fixed` lies on `global`’s zero line for the whole run. No gradient norm reaches the clip threshold.

What the sweep does establish. The specification verbatim, at the $N = 2$ it is motivated by, discards the group structure on 100% of steps (25–30% at $N = 8$); `js_pooled` keeps 56% of it at $N = 2$ and 85% at $N = 8$, the mild shrinkage Stein’s result asks for. The “more stable training” claim finds no support: no arm clipped a single gradient step, sampled-token entropy fell from 0.52 to 0.26–0.33 for every arm alike, and the per-step gradient noise-to-signal ratio (Zeng et al. (2025), eq. 17–18, across micro-batches) is 58–182 with 32–45% of steps returning a non-positive signal estimate. At 8 or 32 prompts per step the gradient is essentially all noise for every baseline, and the ratio cannot separate them. Those steps are reported as right-censored ($+\infty$) rather than dropped, which would bias the median down. Figure 2 shows seed 0 of both shapes.

5.4 A WITHDRAWN MEASUREMENT

Because training runs cannot resolve the effect, an earlier draft measured the estimator directly: every baseline scored the same frozen-policy rollouts, and the trace of the gradient covariance over $\|\mathbb{E}[g]\|^2$ was estimated with Hutchinson probes and bootstrapped over 200 batches. It reported `js_loo` at -21.4% $[-51.5, -7.8]$ against GRPO. Two defects invalidate it: the probes reused random entries across equal-shaped parameter tensors, and every rollout was drawn with the checkpoint’s default repetition penalty of 1.1 rather than from the policy (Appendix C). The instrument was also fragile on its own terms, since $\|\mathbb{E}[g]\|^2$ was 8% of $\mathbb{E}\|g\|^2$ and obtained as a difference. The projection was fixed and a probe-free estimator added — for independent batches $\mathbb{E}\langle g_b, g_{b'} \rangle = \|\mathbb{E}[g]\|^2$ exactly, so disjoint pairs of batches give unbiased samples of both quantities — and both are tested on simulated gradients. The measurement was not repeated on the model within the time budget. The historical table is in Appendix E as a record of the instrument, not as evidence.

6 RELATED WORK

Zeng et al. (2025) is the same idea with its preconditions handled: both the within-group and the between-prompt variance are estimated from the batch, both levels are leave-one-out so the policy gradient stays unbiased, and there are no hyper-parameters. That is `js_loo` here. They report 0.65–1.5 points over GRPO on this model and task at $K = 64$ and 11–67% lower gradient variance. EBPO (Han et al., 2026) takes the empirical-Bayes route: it shrinks toward a running global prior rather than the batch mean, with shrinkage set by the within/between variance ratio. Its stated feature — an all-zero group still receives a non-zero signal — is the mechanism Sec. 3 files under bias. Efron & Morris (1975) applied James–Stein to binomial proportions, which is exactly the 0/1-reward case; their remedy for $\sigma^2 \neq 1$ was an arcsine-square-root variance-stabilising transform before shrinking. We pool the within-group standard deviation instead; the transform is the more orthodox route and is untried here. RLOO (Ahmadian et al., 2024) and Dr.GRPO (Liu et al., 2025) motivate the `rloo` control and the no-scaling default respectively.

7 LIMITATIONS AND CONCLUSION

Three seeds per arm at two group shapes leave every accuracy difference inside its own standard error, and the same-algorithm pair spans 0.5–6.5 points; no accuracy claim is made in either direction. The paired gradient-variance measurement, the one instrument that could separate the baselines at this scale, is withdrawn and was not repeated. `rloo` and `js_loo` were not included in the three-seed sweep. One model, one task; whether the collapse threshold moves with reward density is untested.

What the work does establish is exact where it is exact. The specified estimator, applied to 0/1 rewards as its pseudocode does, is the global batch mean — asymptotically in K for any N , and identically for every $K > 6$ at $N = 2$ — and the training logs show it. Honouring the specification’s own precondition restores the estimator Stein’s theorem describes, with a shrinkage factor that follows the empirical-Bayes closed form. Whether that estimator improves training is a question this scale of experiment cannot answer, and a protocol that reports a single seed, or a t -test with two degrees of freedom, will answer it wrongly.

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A THE SPECIFICATION

GRPO.md gives, in order: the motivation (small- N group means are noisy); the model $X_k \sim \mathcal{N}(\mu_k, V)$ with $V = 1/N$ under the stated assumption that rewards have been rescaled so $\sigma^2 \approx 1$; the positive-part James–Stein estimator of Eq. (1) with $c = (K - 3)/N$; ten implementation steps (group means, grand mean, S , c , $K < 4$ fallback, shrink factor with ε guard, $S = 0$ case, clamp, shrunk means, advantages); a Python listing; and the optional division by the within-group standard deviation.

B CONFORMANCE WITH THE LISTING

`tests/test_spec_conformance.py` transcribes the ten steps and, separately, the Python listing character for character, and runs both against the implementation. Away from the edge cases the listing and `js_fixed` agree to 10^{-12} on continuous and on 0/1 rewards. On its own edge cases the listing does not run:

input	behaviour of the listing
$K < 4$	<code>IndexError: $\bar{X}$ is a 0-d tensor, so <code>unsqueeze(1)</code> is out of range</code>
$S = 0$	<code>TypeError: Python’s <code>max(S, eps)</code> returns the float <code>eps</code> once $S < \varepsilon$, and <code>torch.clamp</code> rejects a float</code>
$K = 8, N = 8, 0/1$ rewards, $S > 0$	runs, and equals <code>js_fixed</code>

Two departures, both asserted. Step 5’s prose returns the per-group means for $K < 4$; the listing, once its indexing is repaired, returns rewards centred on the global mean — the `global` arm, no longer GRPO. We follow the prose. Step 7’s “ $S = 0 \Rightarrow$ shrink factor 1” is unreachable in the listing; we clamp to 0 instead, which is observationally identical since $S = 0$ forces `diff = 0`.

C CORRECTIONS

Found in review after the first draft; all fixed in code.

- Off-policy rollouts (2026-09-04).** `generate()` set temperature, top- p and top- k but not `repetition_penalty`, which fell through to the checkpoint’s `generation_config` — 1.1 for Qwen2.5-Instruct. Completions were sampled from a penalised distribution while the loss used the raw policy’s log-probabilities, in every arm of every experiment. The loop now passes 1.0 explicitly. The three-seed sweeps of Sec. 5.3 were run after this fix; the seed-0 five-arm sweep of Appendix D and the gradient measurement were not.
- Projection bug (2026-09-03).** The Hutchinson probes reused random entries across equal-shaped parameter tensors. Fixed (continuous stream per probe); regression tests compare projected with exact covariance.
- Completion mask (2026-09-04).** Only `<|im_end|>` counted as a stop token; Qwen2.5 also stops on `<|endoftext|>`, which is its pad token, so such a completion was scored as 512 valid tokens including padding. The mask now stops on every terminator plus the pad id.
- Shrinkage diagnostic (2026-09-04).** Groups whose mean equals the grand mean were logged as unshrunk rather than undefined. The synthetic tables were regenerated with the

corrected diagnostic (global now reads its definitional 0; MSE columns moved by under 1.5 points).

D SEED-0 SWEEP UNDER THE 1.1 SAMPLER

The first sweep, superseded, kept for the record: five arms, 150 steps, same seed, same 200 test problems.

arm	eval@0	eval@final	Δ	p	shrink	collapsed
vanilla	42.5%	45.5%	+3.0	0.24	1.000	0/150
js_pooled	42.5%	45.0%	+2.5	0.27	0.850	1/150
js_loo	42.5%	45.5%	+3.0	0.21	0.818	0/150
js_fixed	42.5%	47.0%	+4.5	0.11	0.202	64/150
global	42.5%	47.5%	+5.0	0.06	0.025	133/150

Nothing is significant; the 2.5-point spread across five arms is below the floor one arm shows against itself. The shrinkage column separates into tiers accuracy cannot resolve, and `js_pooled` and `js_loo`, derived independently, land within 0.03 of each other. `global`’s 0.025 is the diagnostic artefact of Appendix C. This sweep’s clip counts (36 of 150 steps for `global` against 9 for `vanilla`) and entropy differences vanish under the corrected sampler.

E THE WITHDRAWN GRADIENT MEASUREMENT

200 paired batches, $K = 8 \times N = 8$, policy frozen, mean accuracy 29.1%, 1.1 sampler, buggy projection. Intervals were a paired bootstrap over batches conditional on 24 probes.

baseline	noise/signal	vs GRPO	
vanilla	11.5	—	
rloo	11.5	−0.3% [−2.5, +6.9]	instrument check
js_loo	9.1	−21.4% [−51.5, −7.8]	withdrawn
js_pooled	9.9	−14.4% [−38.2, +0.8]	withdrawn
js_fixed	6.0	−48.1% [−81.3, −15.8]	biased estimator
global	5.8	−49.7% [−83.5, −15.6]	biased estimator

Even had the numbers stood, `js_fixed` and `global` would not be “best”: they target a different $\mathbb{E}[g]$, since handing an all-correct group a positive advantage inflates $\|\mathbb{E}[g]\|^2$ with a component that merely pushes up easy prompts. A larger denominator is not more signal.

F REPRODUCTION

```
uv sync && python main.py smoke
python -m pytest tests/ -q # 162 tests
python main.py synthetic --k 32 --n 2 4 8 16
python experiments/synthetic_mse.py --k 4 8 16 32 64 --n 8 \
  --trials 3000 --out results/synthetic_k_sweep.json
python main.py sweep --seeds 0 1 2 --k 32 --n 2 --track-grad-var \
  --baselines vanilla js_fixed js_pooled global
python main.py sweep --seeds 0 1 2 --k 8 --n 8 --track-grad-var \
  --baselines vanilla js_fixed js_pooled global
python main.py compare --glob 'train*_k8n8_s*.jsonl'
python main.py compare --glob 'train*_k32n2_s*.jsonl'
python main.py figures
```

The 24 per-step logs of Sec. 5.3 are committed under `results/`; they are not reproducible bit-for-bit (Sec. 5.3), which is why they are kept.